ABSTRACT

This experiment examined how different light wavelengths affect the photosynthesis rate in *Spinacia oleracea* (spinach) by measuring the flotation rate of infiltrated spinach leaf disks in a sodium bicarbonate solution. The disks were exposed to red, orange, green, blue, and violet light using plastic film filters, with white light as a control. The number of floating disks, indicating oxygen production from photosynthesis, was recorded over 30 minutes. Results showed that violet and red light produced the fastest flotation rates, while blue and green light resulted in slower responses. These findings support the hypothesis that certain wavelengths, particularly at the red and violet ends of the spectrum, enhance photosynthetic efficiency in *Spinacia oleracea*. Deviations from the hypothesis, like violet outperforming red, may be due to the intensity or spectral quality of the blue-associated wavelengths. This experiment aligns with existing research and highlights the potential for optimizing artificial lighting in agriculture to maximize photosynthesis. Future studies could improve precision using LED lights and explore cross-species comparisons or direct measurements of glucose and oxygen output.

INTRODUCTION

Photosynthesis is the process by which plants utilize carbon dioxide, water, and light to produce glucose. The growth and development of photosynthetic organisms depend on access to light. Light is absorbed through the plant's chlorophyll and carotenoids, located in the chloroplasts. To harvest light, plants use antenna complexes composed of accessory pigments that absorb specific wavelengths of light. These antenna complexes then transfer the absorbed energy to chlorophyll a, which initiates the electron transport chain. This process produces ATP and NADPH, enabling the light-independent reactions to occur. Therefore, the remainder of the photosynthetic process depends on chlorophyll a's ability to absorb these wavelengths (textbook). Some plants prefer specific wavelengths depending on the composition of their antenna complexes (citation). Several factors can influence these fluctuations, including plant morphology, the chlorophyll a-to-b ratio, carotenoid levels, and other polyphenol and antioxidant contents (citation).

Leafy vegetables are a crucial component of the human diet, and *Spinacia oleracea* is an important winter leafy vegetable that grows close to the ground. It is a commonly harvested plant that undergoes the photosynthetic process. (citation) Chlorophyll and carotenoids exhibit maximum absorption peaks at specific wavelengths, making certain types of light more effective. Many plants, including *Spinacia oleracea*, perform optimally under red and blue light, as chlorophyll a absorbs these wavelengths most efficiently (citation). For example, red light (around 650 nm) is particularly effective in promoting light-dependent reactions and plays a key role in physiological processes such as stem elongation, leaf expansion, and flowering. Meanwhile, blue light influences stomatal conductance, phototropism, and the relocation of chloroplasts under specific conditions (citation).

Research supports the observed results in this experiment. A study from BMC Plant Biology found that red and blue LED light had the most effective pre-treatment role in mitigating

drought stress in *Melissa officinalis*, a plant that undergoes a similar photosynthetic process, by enhancing glucose production (citation). Similarly, a March 2025 study reported that 75% of spinach accessions showed the greatest increase in fresh weight when red light made up 50–75% of the total light composition, indicating more efficient photosynthesis and greater glucose production (citation). In addition, a study published by Horticulturae demonstrated that radish (*Raphanus sativus*) achieved the highest root, shoot, and stem growth under red light exposure (citation). These findings collectively support the conclusion that red and blue wavelengths promote more active photosynthesis across various plant species.

This study aimed to examine the effect of light colour on the rate of oxygen production in *Spinacia oleracea* by measuring how quickly spinach disks rise to the surface of water. *Spinacia oleracea* is commonly used in experiments due to its accessibility and high chlorophyll concentration, which contributes to its green leafy structure. Although various wavelengths are essential for plant function, it is hypothesized that red light will cause the disks to rise the fastest due to greater oxygen production and increased photosynthetic rates, as demonstrated in previous studies. This hypothesis is based on findings that similar leafy green plants, such as *Melissa officinalis*, exhibit the highest photosynthetic activity when exposed to red LEDs. In conditions with green or yellow light, the plant's antenna complex is less effective at absorbing light, resulting in reduced photosynthesis, lower oxygen production, and therefore, slower disk rise rates. Although chlorophyll also absorbs high rates of blue light, numerous studies have shown that blue light displaces chloroplasts and makes photosynthesis much less efficient. (citation)

MATERIALS AND METHOD

This experiment investigated the effect of different light colours on the rate of photosynthesis in *Spinacia oleracea* using a floating leaf disk. A 0.20% sodium bicarbonate solution was prepared by dissolving 0.20 grams of sodium bicarbonate in 100 mL of distilled water. Four drops of liquid dish soap were added to the solution to reduce surface tension and facilitate infiltration into the air spaces of the leaf disks. A single-hole punch was used to prepare uniform disks of spinach, carefully avoiding major leaf veins. Twelve leaf disks were used per trial, and only those prepared immediately before use were selected to ensure consistency and prevent deterioration.

Infiltration was carried out using a 10 mL plastic syringe. The plunger was removed, and the 12 disks were placed inside the barrel. After refitting the plunger, approximately 5 mL of the sodium bicarbonate and soap solution was drawn into the syringe. The syringe was then held with the tip upward to remove any excess air, and a vacuum was created by covering the tip and pulling back on the plunger for 10–15 seconds while gently shaking the syringe. This process was repeated until the majority of the disks sank, indicating successful infiltration. The infiltrated disks and solution were then returned to the beaker for exposure to light.

The experimental setup consisted of placing the beaker containing the infiltrated leaf disks in a clamp on a retort stand, positioned approximately 15 cm above a constant white light source (a lightbulb) to simulate sunlight. To test the effect of different wavelengths, coloured

plastic films (red, blue, green, violet, and yellow) were individually placed over the lightbulb, altering the spectrum of light that reached the beaker. A white light trial was also conducted as the control, representing full-spectrum illumination. The intensity and distance of the light source were kept constant across all trials. Temperature and solution volume were also controlled to isolate light colour as the independent variable.

Figure 1 shows the cup meant to contain the infiltrated spinach leaf disks in a sodium bicarbonate and dish soap solution, held in place by a clamp on a retort stand. A white lightbulb positioned directly beneath the beaker served as the artificial light source, simulating sunlight. A coloured plastic film was placed over the bulb to filter the light and test the effect of different wavelengths on the rate of photosynthesis. The vertical distance between the light source and the beaker was kept constant to ensure equal light intensity across all trials. The entire apparatus was used to measure how various light colours influenced the time it took for the leaf disks to float, indicating oxygen production through photosynthesis.



Figure 1: Experimental Setup for Photosynthesis Light Colour Test

Once the light was turned on, a timer was started. The number of leaf disks that floated to the surface was recorded every minute for 30 minutes, as the rising of the disks indicated oxygen accumulation due to photosynthesis. The endpoint for each trial was the time taken for all 12 disks to rise or the number that had floated by the end of the 30 minutes. Data was recorded in tables and later used to generate graphs comparing the average rate of disk flotation under each light condition.

RESULTS

The number of floating spinach disks increased over time for all light wavelengths tested, with notable differences in the ET_{50} and ET_{100} (**Table 1**). Violet light resulted in the fastest flotation, with all 12 disks floating by 12 minutes. Red and orange light followed closely, reaching 12 floating disks by 16 minutes. Green light showed a steady but slower increase, reaching 12 disks by 20 minutes. The blue light caused the slowest response, reaching 12 disks

by 28 minutes. Overall, more extreme wavelengths (red and violet) led to faster flotation than cooler ones (green and blue).

Table 1: Time of ET50 and ET100 of Spinach Disks under Certain Wavelengths

	Red (650nm)	Orange (610nm)	Green (540nm)	Blue (470nm)	Violet (400nm)
$Time_0$ (mins)	0	0	0	0	0
$Time_{50}$ (mins)	~10	~5.5	12	~18	~6.5
$Time_{100}$ (mins)	16	16	20	27	12

Figure 2 illustrates the rate of photosynthesis, represented by 1/ET50, plotted against light wavelength (in nm). The graph shows a non-linear relationship, with the rate of photosynthesis varying by wavelength. It is observed that the rate is highest under red, orange, and purple light, as indicated by the peaks in the graph at those wavelengths. In contrast, the rate is lowest under green and blue light, where the curve dips, suggesting that these wavelengths are less effective for photosynthesis.

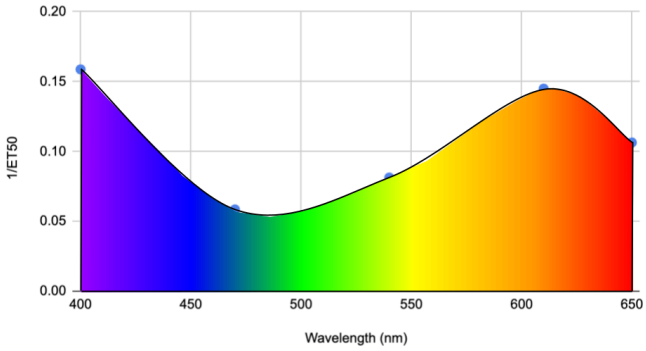


Figure 2: 1/ET50 of Spinacia oleracea under Wavelengths of Light

Figures 3 through **7** display the ET_{50} (Effective Time for 50% of the spinach disks to float) for each of the six light conditions tested: red (650 nm), orange (610 nm), green (540 nm), blue (470 nm), violet (400 nm), and control/no light. Each graph plots the number of floating

disks over time, allowing for the visual identification of the point at which half (6 out of 12) of the disks floated, representing the relative rate of photosynthesis. The curves show that disks exposed to violet, orange, and red light reached ET_{50} the fastest, while green and blue light produced slower ET_{50} values. These visual trends further support the numerical data indicating that light wavelength significantly affects photosynthetic activity in spinach leaves.

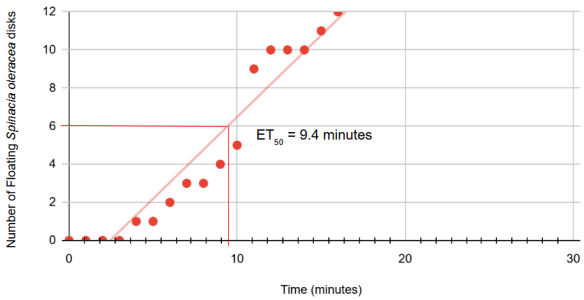


Figure 3: ET_{50} of *Spinacia oleracea* under Red Light (650 nm)

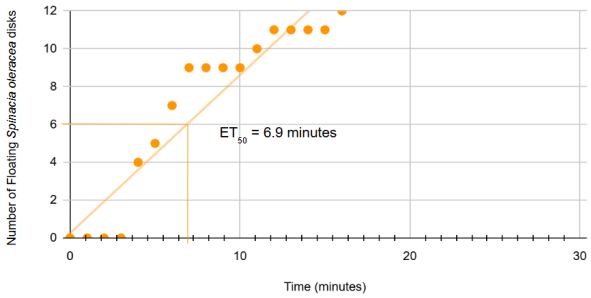


Figure 4: ET_{50} of *Spinacia oleracea* under Orange Light (610 nm)

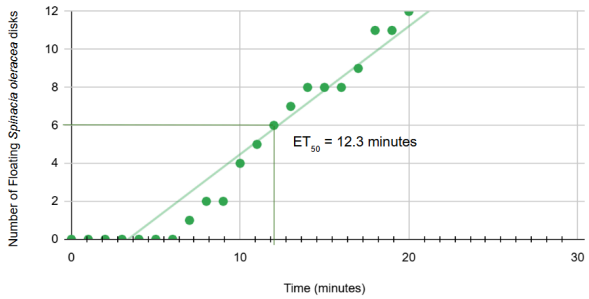


Figure 5: ET_{50} of *Spinacia oleracea* under Green Light (540 nm)

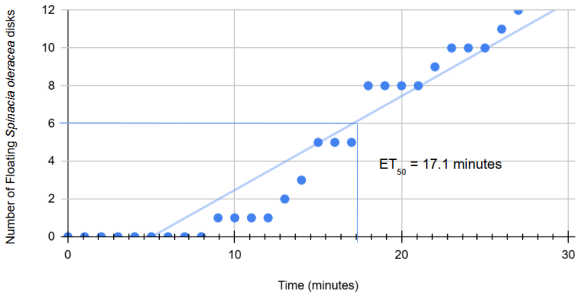


Figure 5: ET_{50} of *Spinacia oleracea* under Blue Light (470 nm)

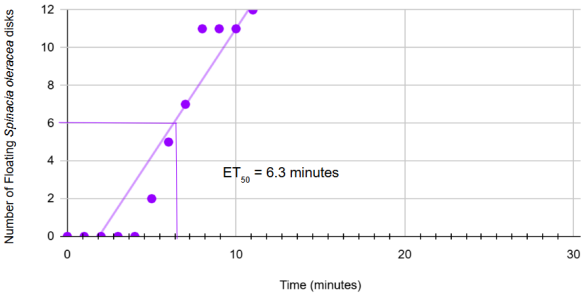


Figure 6: ET_{50} of *Spinacia oleracea* under Violet Light (400 nm)

DISCUSSION

By understanding how fast the spinach disks floated in the solution, we can extrapolate how much oxygen was produced. The main two results of photosynthesis are glucose and oxygen. The glucose would be used by the plant and the oxygen would be expelled via the leaf's stomatas (textbook). The gas production would thereby cause the disks to float. Therefore, it can be concluded that the faster the disks rose, the more oxygen was produced, meaning a higher rate of photosynthesis. Therefore, these results can be used to make conclusions about photosynthesis rate, meaning that amongst all of these wavelengths, violet and red wavelengths have the most optimal rates of photosynthesis for the plant. The purpose of this experiment was to investigate how different wavelengths of light affect the rate of photosynthesis, as measured by the number of floating spinach disks over time.

Originally, it was hypothesized that red light would result in the fastest rate of photosynthesis, however, results indicated that violet light resulted in a faster rate of photosynthesis compared to red. This most likely occurred as a result of the wavelengths in the violet pigment being much closer to the blue pigment, explaining the slow blue results as well. It is possible that the concentration of high blue light impacted the chloroplast negatively. Studies have shown that intense blue light promotes the dislocation of the chloroplasts in the cell (citation). As these are the sites of photosynthesis for the plant, it is possible that the relocation negatively impacted the plant's ability to properly absorb the wavelengths of light and therefore resulted in less oxygen production, hence less photosynthesis. All colours showcased a positive linear correlation, meaning that photosynthesis for all wavelengths increased as time went on.

All studies discuss red and blue wavelengths, but there is minimal discussion about violet or orange wavelengths. One source of possible error is the fact that the violet wavelength was much closer to the desired deep blue wavelength compared to the blue pigment, that was much closer to green/turquoise. As plants are notoriously weak at absorbing green wavelengths, this pattern would make sense.

From this experiment, an action spectrum can be created to visualize the wavelengths that are favoured in *Spinacia oleracea*. An action spectrum plots the rate of photosynthesis relative to the wavelength of light present. As seen in **Figure 8**, photosynthesis peaks at violet and red wavelengths, while experiencing a severe dip in blue and green wavelengths.

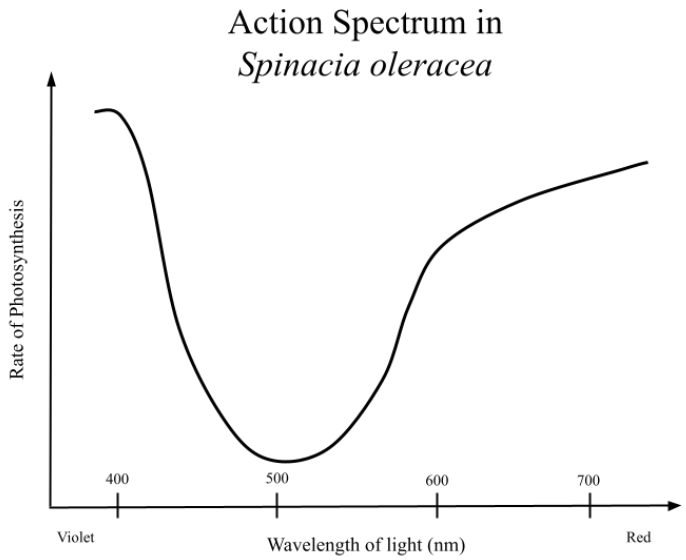


Figure 8: Action Spectrum in *Spinacia oleracea* from Experiment Results

Studies have shown that chlorophyll a and b, the primary pigments involved in photosynthesis (the primary electro, absorb red and blue/violet wavelengths most effectively while reflecting green light, which is why plants appear green. This supports the finding that purple (400 nm) and red (650 nm) light resulted in faster spinach disk flotation, a proxy for higher photosynthetic rates. Similar results have been reported in academic studies and experiments, such as those published in plant biology journals or high school biology labs, where red and blue light consistently outperform green in driving photosynthesis.

In one experiment, researchers found that lettuce grown under red and blue LED lights showed significantly more biomass accumulation than under green light (citation). In this study, scientists used their knowledge about light to make advancements in the fields of vertical modern agriculture. Maximizing agricultural efficiency means ensuring that light absorption for plants is optimized in order to maximize the rate of photosynthesis occurring. This reinforces the idea that the experiment is in line with real-world applications, especially in controlled-environment agriculture.

Several potential sources of error may have affected the reliability of the results. First, the trials for different wavelengths were conducted on different days due to time constraints, which could introduce variability in environmental conditions such as temperature, light intensity, or the freshness of the spinach leaves. Although most disks were taken from the same bunch of spinach, some disks were cut out from other leaves due to the experiment taking place across three days. These uncontrolled factors may have influenced the rate of photosynthesis independently of light wavelength. Additionally, the colored film used to filter light may not have been consistent across trials. It's possible that some films were partially detached or did not fully block out other wavelengths, reducing the accuracy of the light filtering and allowing mixed light to affect the results. Furthermore, if the film used was not of high optical quality, it may have failed to isolate the intended wavelengths effectively. These issues could lead to

inaccuracies in measuring the true effect of each specific light wavelength on the rate of photosynthesis.

Future investigations could expand on this experiment by testing the effect of light wavelength on photosynthesis in different plant species, such as aquatic plants, succulents, or shade-tolerant varieties, to see if similar trends are observed. It would also be valuable to examine the impact of light intensity in combination with wavelength, as higher or lower intensities may amplify or diminish the rate of photosynthesis. Additionally, conducting trials using LED lights with precise wavelength outputs instead of colored films could improve accuracy and control. Exploring how other pigments (such as anthocyanins or carotenoids) influence light absorption across different wavelengths could also deepen the investigation. Finally, extending the experiment to measure oxygen output or glucose production directly would provide a more quantitative understanding of photosynthetic efficiency under different light conditions.

These results have valuable applications in agriculture, particularly in optimizing crop yields in both traditional and controlled environments. Understanding which wavelengths of light most efficiently drive photosynthesis can help farmers and agricultural engineers design more effective lighting systems for greenhouses and vertical farms. For example, using red and violet/blue LED lights could enhance plant growth while reducing energy costs compared to using full-spectrum lighting. As agriculture continues to expand globally and adapt to climate challenges, leveraging photosynthetic efficiency through light management could contribute to sustainable food production, resource conservation, and increased crop productivity.

REFERENCES